

Aftermath : AI-Human asymmetry in VR Interaction

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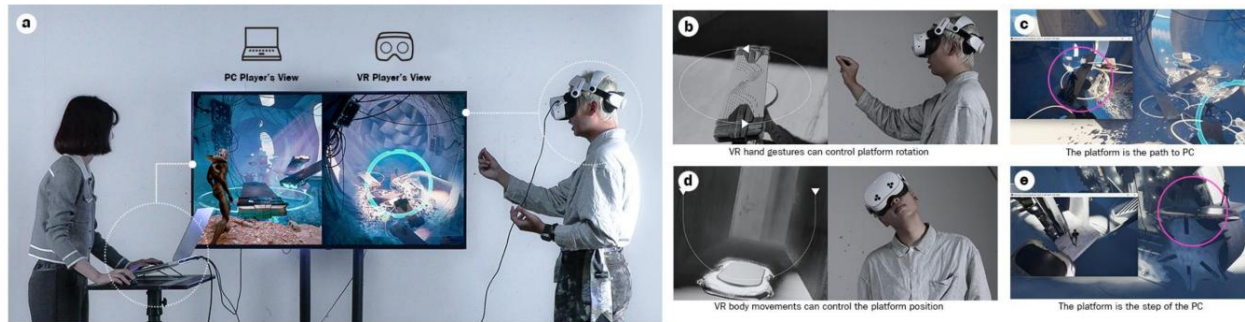


Figure 1: (a) VR-PC asymmetric gameplay interface, (b) Hand gestures control platform rotation, (c) Platforms serve as dynamic paths for the PC player, (d) Head tilts shift platform positions, (e) Platforms function as PC traversal steps.

ABSTRACT

An asymmetric VR experience interrogating the unequal dynamics of control between human and AI systems, revealing the quiet hierarchies embedded in our interactions with technology. *Aftermath* foregrounds the asymmetrical power dynamics between human ingenuity and AI omniscience, probing whether artificial systems can truly understand, replicate, or ultimately surpass human creativity. Game mechanics are designed to simulate AI's control over human agency in subtle yet powerful ways. The asymmetry between the VR player (AI) and the PC player (human) mirrors the real-world dynamic where AI operates from an unseen, omnipresent perspective, shaping human choices without direct intervention. The VR player can manipulate the environment in real-time, controlling how the PC player perceives and interacts with space. The PC player, in contrast, must navigate this dynamic world, always reliant on AI's assistance. Through this unequal relationship, *Aftermath* invites players to confront the evolving balance between human agency and artificial influence.

Index terms: Virtual Reality, Interactive narratives, Speculative design, AI.

1 INTRODUCTION

As AI systems become increasingly embedded in our social, creative, and infrastructural frameworks, new asymmetries are

emerging between human and machine capabilities. These asymmetries, particularly in perception, information processing, and control—are no longer abstract concepts but lived experiences mediated through technology. This project investigates how such imbalances could redefine interaction paradigms between humans and AI, not through static analysis but via embodied, participatory experimentation.

Aftermath is an innovative asymmetrical VR-PC cooperative game that transforms speculative design into an interactive research methodology. By assigning players radically different roles—one as a human explorer on PC, the other as an omnipresent AI in VR—the game simulates a future in which artificial systems guide, manipulate, and co-create with human agents in layered, sometimes opaque ways. This dynamic is not merely mechanical; it is psychological, ethical, and spatial.

The VR player interacts with the environment using intuitive gesture-based controls (such as hand rotations to move platforms, or head tilts to shift architectural elements), emphasizing embodied influence without direct physical presence. The PC player, in contrast, engages through exploration and puzzle solving within abandoned architectural ruins that symbolize lost creative agency. The physical disconnection between roles foregrounds an emerging question in human-AI collaboration: can systems that do not share our embodiment or experience still act as collaborators—or do they inevitably become subtle governors?

This research positions *Aftermath* as a critical and technical

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prototype that examines the interplay between control, assistance, and manipulation in future human-AI relationships. By embedding philosophical speculation within an immersive, interactive structure, the project contributes to fields such as XR interaction design, narrative environments, and speculative HCI, offering new pathways to prototype, experience, and evaluating post-anthropocentric futures.

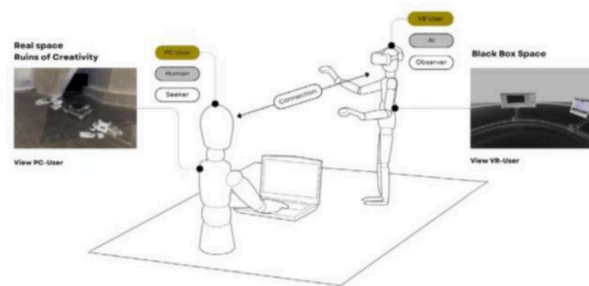


Figure2: Interaction of components for the asymmetric VR framework.

2 VR INTERACTION DESIGN

2.1 GAME MECHANISM

Aftermath explores human-AI power asymmetry through its core game mechanic: asymmetric cooperation between a VR player and a PC player. This interaction is not only technical but symbolic—embodying AI as a disembodied force with expansive control over spatial systems, and humans as grounded agents with limited but meaningful agency.

In the VR role, players control the environment through full-body gesture-based inputs. Three interaction types simulate different levels of environmental influence: first, hand rotation allows manipulation of floating platforms' orientation; second, head tilts adjust the relative positioning of modular elements in real time; and third, nodding movements modulate spatial disc rotation, directly altering the platform trajectory. These embodied gestures mirror how future AI might interface with environments invisibly, guiding human movement without direct contact—an omnipresent but intangible presence.

2.2 LEVEL DESIGN TO EMPHASIZE THIS POWER DYNAMIC

To emphasize this power dynamic, the game's levels are designed with distinct spatial hierarchies and symbolic architecture.

Level One introduces a circular traversal environment. The PC player begins in a fragmented outer ring and must cross hovering, shifting platforms toward a sealed gateway at the heart of the Creativity Ruins. The VR player governs the kinetic behavior of each element—activating, rotating, or disabling platforms—thereby controlling the rhythm and risk of human movement. The space is haunting and monumental, a decaying relic of an era when creativity and AI once coexisted. The ruinous aesthetic contrasts memory and mechanism: soft organic decay meets rigid architectural symmetry, echoing the absence of emotion in a system once meant to enhance imagination.

Level Two transforms the spatial axis. Here, the environment shifts vertically. The PC player must ascend a vast vertical

chamber filled with mechanical contraptions—interlocking gears, suspended pulleys, and fragmented stairs—resembling an ancient timekeeping mechanism. This "clockwork cathedral" symbolizes the total efficiency and cold precision of post-creative AI logic. The VR player orchestrates the entire system from above, rotating gears and shifting mechanisms to assist or hinder the PC player's vertical progress. While the PC player is physically bounded and navigates uncertainty, the VR player's god-like perspective introduces a tension between empowerment and dependence. vero eros et accumsan et iusto odio dignissim qui blandit praesent luptatum zzril delenit augue dui dolore te feugait nulla facilisi.

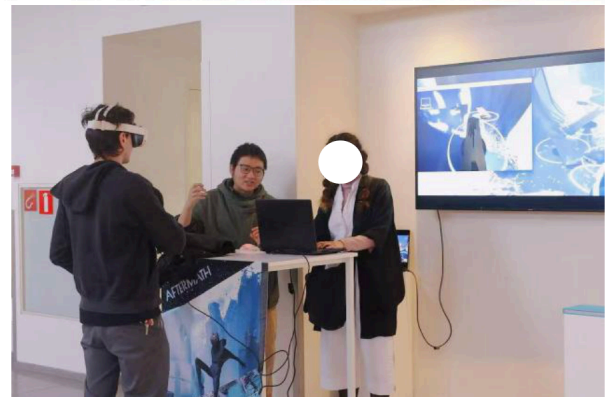
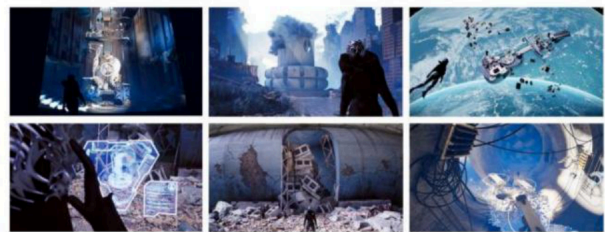


Figure3: VR Gaming environment metaphor for asymmetry between humans and AI.

3 CONCLUSION

Aftermath creates a spatial metaphor for the future relationship between humans and AI. The interaction design forces players to consider questions of trust, control, and autonomy—what it means to collaborate with a system that knows more, sees more, and controls more, but does not feel.

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